Weight-Bearing Feedback Device Design Statement

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There are three main parts to this device:

1) the sensor
2) the electronics/circuits that process the signal
3) the method of reporting to the user.

Here is how we will deal with the parts:

1) Together the heel and ball of foot capture nearly 100% of the ground reaction force during walking. More accurate picture of total load. So whatever sensor method gets chosen, it **must** cover both the ball and heel of the foot.

2) The signal from the sensor must be continuously read by a small microcontroller. The microcontroller converts the raw sensor signal into a percentage of body weight in real time using the body weight the user enters in the beginning. It will be powered by a small portable battery. The battery must be able to last a while.

3) This device will come with a smart phone app. The app will have the percent number on the screen and track it over time. The user gets to decide how the method is reported in real time. For example, the app will have a setting in which the phone could buzz if they go over their limit, or the phone could make a sound. There will be no speaker or light on the shoe, just on the phone.

Using the design statement, we can make three proposed devices:

First device: 1) Conductive foam across the entire foot (the foam is the insole). 2) ESP32 microcontroller with cr2032 battery, both really small, can fit on bottom of insole. 3) iPhone app, with live %, day tracking, buzz/sound alert (user gets to choose).

Second device: 1) 2 FSR one on ball of foot one on heel of foot. 2) Wheatstone bridge circuit + ESP32 microcontroller + 9V battery embedded in the arch. Even though 9V battery is bigger than cr2032, this device needs more power for the Wheatstone bridge. 3) iPhone app, with live %, day tracking, buzz/sound alert (user gets to choose).

Third device: 1) Conductive ink painted between the layers of a standard Dr. Scholl's Heavy Duty insole. Copper tape contacts on either side of the ink layer create the sandwich sensor. 2) nRF52840 microcontroller (built in bluetooth low energy, can last weeks with small battery) with cr2032 battery. 3) iPhone app, with live %, day tracking, buzz/sound alert (user gets to choose).

Decision Matrix for Sensing device:

SENSING DEVICE	Cost	Can you actually built/program/modify the tech or method	How closely does it fit your design requirements	Total
Conductive foam	9	8	10	27
FSR	7	9	8	24
Conductive ink	4	5	10	19

Decision Matrix for Electronics/Circuits:

Electronics/Circuits	Cost	Can you actually built/program/modify the tech or method	How closely does it fit your design requirements	Total
ESP32 microcontroller	7	9	10	26
cr2032 battery	10	9	2	21
Wheatstone bridge circuit	7	3	10	20
9V battery	9	8	10	27
nRF52840 microcontroller	5	6	10	21

Decision Matrix for Method of Reporting

Method of Reporting to User	Cost	Can you actually built/program/modify the tech or method	How closely does it fit your design requirements	Total
Iphone app	2	10	10	22
Light on shoe	10	6	2	18

Now, adding up the points for each proposed idea:

First Device: 27 + 26 + 21 + 22 = 96

Second Device: 24 + 20 + 26 + 27 + 22 = 119

Third Device: 19 + 21 + 21 + 22 = 83

The winner was the 2nd device, however the 2nd device had 5 components, not 4 like the other 2 proposed devices. So, let's normalize the scores. Divide each total by the number of components to get an average score per component:

First Device: 96/4 = 24

Second Device: 119/5 = 23.8

Third Device: 83/4 = 20.75

Now, using the averaged scores, the first proposed device is the winner.